



Genomic diversity and molecular epidemiology of a multidrug-resistant *Pseudomonas aeruginosa* DMC30b isolated from a hospitalized burn patient in Bangladesh

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ABSTRACT

Objectives: *Pseudomonas aeruginosa* is a key opportunistic pathogen causing a wide range of community- and hospital-acquired infections in immunocompromised or catheterized patients. Here, we report the complete genome sequence of a multidrug-resistant (MDR) *P. aeruginosa* DMC30b to elucidate the genetic diversity, molecular epidemiology, and underlying mechanisms for antimicrobial resistance and virulence. **Methods:** *P. aeruginosa* DMC30b was isolated from septic wound swab of a severe burn patient. Whole-genome sequencing was performed under Ion Torrent platform. The genome was assembled using the SPAdes v. 3.12.01 in an integrated Genome Analysis Platform for Ion Torrent sequence data. The genome was annotated using the National Center for Biotechnology Information (NCBI) Prokaryotic Genome Annotation Pipeline. *In-silico* predictions of antimicrobial resistance genes, virulence factor genes, and metabolic functional potentials were performed using different curated bioinformatics tools.

Results: *P. aeruginosa* DMC30b was found as a MDR strain and belonged to sequence type 244 (ST244). The complete genome size is 6 994 756 bp with a coverage of 76.76x, guanine-cytosine content of 65.7% and a Benchmarking Universal Single-Copy Orthologs score of 100. The genome of *P. aeruginosa* DMC30b harboured two predicted plasmid replicons (e.g. IncP-6; 78 007 bp and ColRNAI; 9359 bp), 35 resistomes (antimicrobial resistance genes) conferring resistance to 18 different antibiotics (including four beta-lactam classes), and 214 virulence factor genes. It was identified as the 167th ST244 strain among ~5800 whole-genome sequences of *P. aeruginosa* available in the NCBI database.

Conclusion: The MDR *P. aeruginosa* DMC30b was identified as the 167th ST244 complete genome to be submitted to the NCBI, and the first ST244 isolate sequenced from Bangladesh. The complete genome data with high genetic diversity and underlying mechanisms for antimicrobial resistance and virulence of *P. aeruginosa* DMC30b will aid in understanding the evolution and phylogeny of such high-risk clones and provide a solid basis for further research on MDR or extensively drug resistant strains.

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1. Introduction

Pseudomonas aeruginosa is an opportunistic Gram-negative, non-fermenting bacterial pathogen that causes a wide array of life-threatening, acute and chronic infections in hospitalized and

immunocompromised patients, especially in the intensive care unit (ICU) [1]. It is well established that bacterial proliferation in wounds contributes to infection and delayed wound healing [2]. The microenvironment of chronic wounds is ideal for bioburden and usually contains multiple bacterial species [3]. *P. aeruginosa* is one of the most common bacteria isolated from chronic wounds, and it is highly prevalent in chronic wounds, estimated to be present in about 25% of all cases [4]. Infections caused by *P. aeruginosa* are usually difficult to treat and persistent because of the characteristic high frequency of emergence of multidrug resistance

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(MDR) or extensively drug-resistant (XDR) strains [5,6]. The World Health Organization (WHO) and Centers for Disease Control and Prevention (CDC) have designated *P. aeruginosa* as one of the major (critical) pathogens for which new antibiotics are desperately needed [7]. Moreover, *P. aeruginosa* is a persistent and difficult-to-treat pathogen in many patients and possesses a versatile arsenal of antimicrobial resistance determinants and virulence factors that enable survival, adaptation, and consequent persistence within the complex milieu of infections [8]. The increasingly frequent infections caused by MDR and XDR strains of *Pseudomonas* spp. with limited therapeutic options are associated with high morbidity and mortality worldwide [8]. In particular, the emergence in recent years of MDR and XDR strains due to bacterial expression of resistance genes such as β -lactamases, 16S rRNA methylases, and carbapenemases has led to severe infections with serious global threats to human health, emphasizing the need for novel (antibiotics-independent) treatment strategies. The most troublesome acquired resistance of *P. aeruginosa* is the production of carbapenemases, which confer resistance to most commercially available β -lactams [6]. The presence of carbapenemases in high-risk clones and sequence types (STs) on complete genome, plasmid, or the chromosome of *P. aeruginosa* may be the cause of its successful dissemination in Bangladesh and beyond. Recent molecular and genomic studies reported that most of the infectious and virulent clones of *P. aeruginosa* belonged to ST235, ST244, ST308, ST1006, and ST1060 [9,10].

Recently, increasing prevalence of MDR and XDR strains of the *P. aeruginosa*, with rates of between 15% and 30% in some geographical areas, has been reported [6,10]. Microbial and host factors furthermore affect the ongoing adaptive response exhibited by MDR, XDR, and mutating strains of *P. aeruginosa* in immunocompromised and hospitalized patients. Furthermore, acquisition of advantageous attributes such as virulence factor assembly, motility, antibiotic resistance, and metabolic adaptation of the strains are the major contributors to morbidity and mortality [8]. Several molecular and typing methods have been used to study the evolution and genetic heterogeneity of *P. aeruginosa* because it is characterized by high genetic diversity [8,11]. In recent years, whole genome sequencing (WGS) and RNAseq analysis has enabled the study of isolates collected sequentially from patients and the characterization of the molecular epidemiology and evolutionary patterns and/or trajectories that comprise the hallmark complexity of diverse subclones of *P. aeruginosa* in clinically infected patients [9,10]. The study of molecular epidemiology and genetic diversity in MDR *P. aeruginosa* that harbour genes conferring resistance to a wide range of antibiotics and virulence factors is key to understanding its role in the dissemination of such resistance determinants among clinical and environmental isolates [9,10]. The International *Pseudomonas aeruginosa* Consortium was formed with the aim of genome sequencing > 1000 *P. aeruginosa* genomes, and constructing an analysis pipeline for the study of *P. aeruginosa* evolution, virulence, and antibiotic resistance [12]. Genomics data of this consortium will support molecular epidemiology for the surveillance of outbreaks and has the potential for future genotypic antimicrobial susceptibility testing as well as the identification of novel therapeutic targets and prognostic markers [9,11]. In this article, we described the epidemiological distribution, host range, genomic diversities, antimicrobial resistance genes (ARGs), and virulence factors found in an MDR *P. aeruginosa* DMC30b strain isolated from septic wound swab of a severe burn patient hospitalized in the Dhaka Medical College (DMC), Bangladesh. To elucidate the genetic diversity, molecular epidemiology, and underlying mechanisms for antimicrobial resistance and virulence of *P. aeruginosa* DMC30b, high throughput WGS and downstream bioinformatic analysis were performed.

2. Materials and Methods

2.1. Isolate retrieval and antimicrobial susceptibility tests

P. aeruginosa DMC30b, previously isolated from septic wound swab of a severe burn patient [6], was retrieved and screened for antimicrobial susceptibility tests, genomic diversity, and molecular epidemiological analysis. The isolate was plated onto the Mueller Hinton agar (MHA) (Oxoid, England) and incubated at 37°C for 18–24 h, followed by retrieval of pure colonies. The isolate was tested against a series of antibiotics recommended in the Clinical and Laboratory Standards Institute (CLSI) document M1007 [13] using the Kirby–Bauer method. Twenty antibiotics of 11 antibiotic groups were tested: imipenem (10 mg), meropenem (10 mg), doripenem (10 mg), chloramphenicol (30 mg), ampicillin (10 mg), doxycycline (30 mg), nitrofurantoin (300 mg), gentamicin (10 mg), trimethoprim (5 mg), tetracycline (30 mg), cefalexin (first generation; 30 mg), cefuroxime (second generation; 30 mg), cefotaxime (third generation; 30 mg), cefepime (fourth generation; 30 mg), nalidixic acid (first generation; 30 mg), ciprofloxacin (second generation; 5 mg), levofloxacin (third generation; 5 mg), aztreonam (30 mg), polymyxin B (10 mg) and colistin (10 mg). MIC₉₀s of imipenem, meropenem, ciprofloxacin, chloramphenicol, ceftriaxone, erythromycin, and tetracycline were determined by broth microdilution using 2-fold dilution in the range 2–512 mg/L under incubation at 37°C.

2.2. DNA extraction and whole genome sequencing for *P. aeruginosa* DMC30b

The genomic DNA from *P. aeruginosa* DMC30b was extracted from overnight culture by boiled DNA extraction method using commercial DNA extraction kit, QIAamp DNA Mini Kit (QIAGEN, Hilden, Germany). The quality and quantity of the extracted DNA were measured using a NanoDrop ND-2000 spectrophotometer (Thermo Fisher Scientific, Waltham, MA). DNA extracts with A260/280 and A260/230 ratios of ~1.80 and 2.00 to 2.20, respectively, were considered as high-purity DNA samples [14]. Finally, the harvested DNA was visualized on 1% (w/v) agarose gel and sent for DNA sequencing based on their high purity and adequate concentration.

WGS was done under Ion Torrent platform using 400 bp read chemistry [15]. The sequence reactions were performed using the BigDye terminator cycle sequencing kit v3.1 (Applied Biosystems), purified using the BigDye XTerminator purification kit (Applied Biosystems), and then loaded onto a SeqStudio genetic analyzer capillary sequencer (Applied Biosystems) according to the manufacturer's instructions.

2.3. Genome analysis of *P. aeruginosa* strain DMC30b

The Ion Torrent platform generated FASTQ reads quality was initially assessed by the FastQC tool [16] followed by trimming of low-quality sequences using the Trimmomatic [17], while the quality cut-off value was Phred-20. Trimmed reads were assembled *de novo* using SPAdes v. 3.12.1 [18] in an integrated Genome Analysis Platform for Ion Torrent sequence data (<http://iongap.hpc.iter.es/iongap/newproject>). Coverage of the genomes ranged from 76.76x. The circular visualization of the *P. aeruginosa* DMC30b genome was performed using CGViewer [19]. *P. aeruginosa* DMC30b genome was compared with another complete genome sequence of NCBI reference strain; *P. aeruginosa* PAO1 (NCBI sequence accession number NC_002516 under BioProject: PRJNA57945 and Assembly: GCF_000006765.1) [20]. KmerFinder 3.1 [21] and PathogenFinder 1.1 [22] open-source tools were utilized to identify the species (i.e. *P. aeruginosa*) and pathogenicity of the isolate (DMC30b).

The annotated WGS data were used for sequence typing, antibiotic resistance genes (ARGs) prediction, virulence factor genes (VFGs) profiling, plasmids identification, and metabolic functional analysis. PROKKA Annotation v.3.2.1 [23] was used to predict the function and identification of assembled sequences against the nucleotide and protein sequence database. Benchmarking Universal Single-Copy Orthologs (BUSCO) v.4.1.2 with the 'bacteria_odb10' data set was used to measure the completeness of the genome [24]. The genetic relatedness of the predicted sequence types, core genome multilocus sequence typing analysis (cgMLST), and bacterial source tracking of the *P. aeruginosa* DMC30b and all 166 ST244 *P. aeruginosa* isolates currently deposited in NCBI GenBank database were performed through BacWGSTdb 2.0 server [25]. The Comprehensive Antibiotic Resistance Database (CARD) [26], and VirulenceFinder [27] databases were used to detect ARGs and VFGs, respectively. The CARD (<https://card.mcmaster.ca>) integrated within the AMR++ pipeline identified the respective genes or protein families coding for the ARGs markers in the *P. aeruginosa* DMC30b strain [26]. We utilized SnapGene Viewer web tool (<https://www.snapgene.com/snapgene-viewer/>) to visualize the virulence plasmid of the *P. aeruginosa* DMC30b. Annotation of the genome was also performed with the Rapid Annotation using Subsystems Technology (RAST) server (<http://rast.nmpdr.org/>) to detect the genomic functional features.

3. Results and Discussion

3.1. Genome annotation of *P. aeruginosa* strain DMC30b

The KmerFinder 3.1 detected the DMC30b isolate as *P. aeruginosa* DMC30b, while the pathogenicity of the isolate was confirmed (0.973 out of 1.00, close to the pick value, indicating higher pathogenicity) by PathogenFinder 1.1. The *de novo* genome assembly revealed that the *P. aeruginosa* DMC30b genome was 6 994 756 bp with a coverage of 76.76x. The draft genome had 4 595 555 reads assigning for guanine-cytosine content of 65.7% (Table 1). Analysis of the annotated genome using CG view produced a circular genome map illustrating the GC content, tRNA, rRNA, coding sequences (CDS), and contigs (Fig. 1). The genome of *P. aeruginosa* DMC30b possessed 474 contigs including N50 contig 56 583 bp, L50 contig 35 bp, and two plasmids. Of the detected contigs, the longest and shortest contig size were 252 681 bp and 202 bp, respectively. Notably, preliminary sequence analysis revealed 44 insertion sequences, 21 predicated genomic islands, six prophage-related sequences and two CRISPR (clustered regularly interspaced short palindromic repeats) arrays. The genome of *P. aeruginosa* DMC30b encoded for 7036 genes (Table 1). A physical genome map of *P. aeruginosa* DMC30b in comparison to another reference strain *P. aeruginosa* PAO1 is shown in Fig. 1. PROKKA annotated genome using PATRIC (pathosystems resource integration center) produced a circular genome map (Fig. 2), which identified various gene features, 6685 protein CDSs, 67 RNAs including 4 rRNAs (5S = 1, 16S = 1, 23S = 2), 59 tRNAs, 4 ncRNAs, and 285 pseudogenes. The protein features consist of 3060 hypothetical proteins, 2379 functional assignments, and 1421 gene ontology (GO) assignments. Genome completeness analysis with BUSCO showed the presence of 100% complete BUSCOs in the hybrid assembly.

The diverse and dynamic genetic composition of *P. aeruginosa* enables this bacterium to colonise various environments, including in humans, where it can cause opportunistic infections [28]. The genomic features of the *P. aeruginosa* DMC30b genome described in this study are corroborated with the genome composition of different *P. aeruginosa* strains reported previously from other countries [10,29,30]. For instance, the *P. aeruginosa* DMC30b showed 99.51% nucleotide identity across the entire genome with one of the most commonly used reference strains; *P. aeruginosa* PAO1 [31].

Table 1

Pseudomonas aeruginosa strain DMC30b genome characteristics.

Attribute(s)	Value	% of Total ^a
Genome size (bp)	6 994 756	100.00
DNA coding (bp)	6 470 148	92.50
Number of reads assemblies	2 550 071	100
Genome coverage	76.76x	76.76
DNA G + C (bp)	4 595 555	65.7
Total contigs	474	
Longest contig size (bp)	252 681	3.61
Shortest contig size (bp)	202	0.002
Contig N50 (bp)	56 583	0.81
Contig L50 (bp)	35	
Total genes	7036	100.00
Coding sequences (CDSs)	6969	99.04
CDSs (Protein coding)	6685	95.01
RNA genes	67	100
tRNA genes	59	88
rRNA genes	4	6
ncRNA genes	4	6
Pseudo genes	285	4.05
Genes with function prediction	7146	100
Genes assigned to SEED subsystems	1856	26
Number of subsystems	340	
CRISPR arrays	2	
Number of plasmids	2	

CRISPR, Clustered Regularly Interspaced Short Palindromic Repeats.

SEED (<http://www.theseed.org/>) is a web-based public database into a single resource. The SEED contains all publicly available genome sequences. The database contains accurate and up-to-date annotations based on the subsystems concept that leverages clustering between genomes and other clues to accurately and efficiently annotate microbial genomes. It also includes functional annotations, subsystems, and GO (gene ontology) terms for proteins in all microbial genomes.

^a Total based on either the size of the genome in base pairs (bp) or the total number of genes in the annotated genome. N50: the sequence length of the shortest contig at 50% of the total genome length; L50: count of smallest number of contigs whose length sum makes at least 50% the total genome.

The prevalence of MDR or extensively drug resistant strains of *P. aeruginosa* reduces treatment options, significantly increasing morbidity rates. *P. aeruginosa* PAO1 is a spontaneous chloramphenicol-resistant mutant of the original PAO strain (earlier called "*P. aeruginosa* strain 1") that had been isolated in 1954 from a wound in Melbourne, Australia [32]. Over decades, discordant phenotypes of *P. aeruginosa* PAO1 sublines (e.g., *P. aeruginosa* DMC30b) have emerged that show an ongoing microevolution of genotype and phenotype that jeopardizes the reproducibility of research.

3.2. Molecular epidemiology of *P. aeruginosa* strain DMC30b

The *P. aeruginosa* DMC30b belonged to ST244 according to Achtman's MLST scheme for *P. aeruginosa*. To determine the genomic epidemiological characteristics of *P. aeruginosa* strain DMC30b in a global context, the phylogenetic relationships between *P. aeruginosa* DMC30b and a total of 166 other ST244 *P. aeruginosa* strains currently deposited in the NCBI GenBank database were analysed (Fig. 3). The cgMLST analysis revealed that the closest relatives of the *P. aeruginosa* DMC30b genome were one carbapenem-resistant ST244 *P. aeruginosa* clone in burn patients in Yunnan province, China [10], and another ST244 strain, *P. aeruginosa* MRSN 17623, carrying the blaVIM-6 gene recovered from human clinical samples in the USA (<https://www.pseudomonas.com/strain/show/2623>), which differed by 285 cgMLST loci (Fig. 3). For the first time, ST244 is reported to emerge in burn patients in Bangladesh, and it exhibited a higher level of resistance to several antibiotics. These results highlight the need for surveillance of MDR-resistant isolates in burn patients [10]. Epidemiological source tracking revealed that the *P. aeruginosa* DMC30b strain has a close evolutionary relationship with many strains of *P. aeruginosa* originating

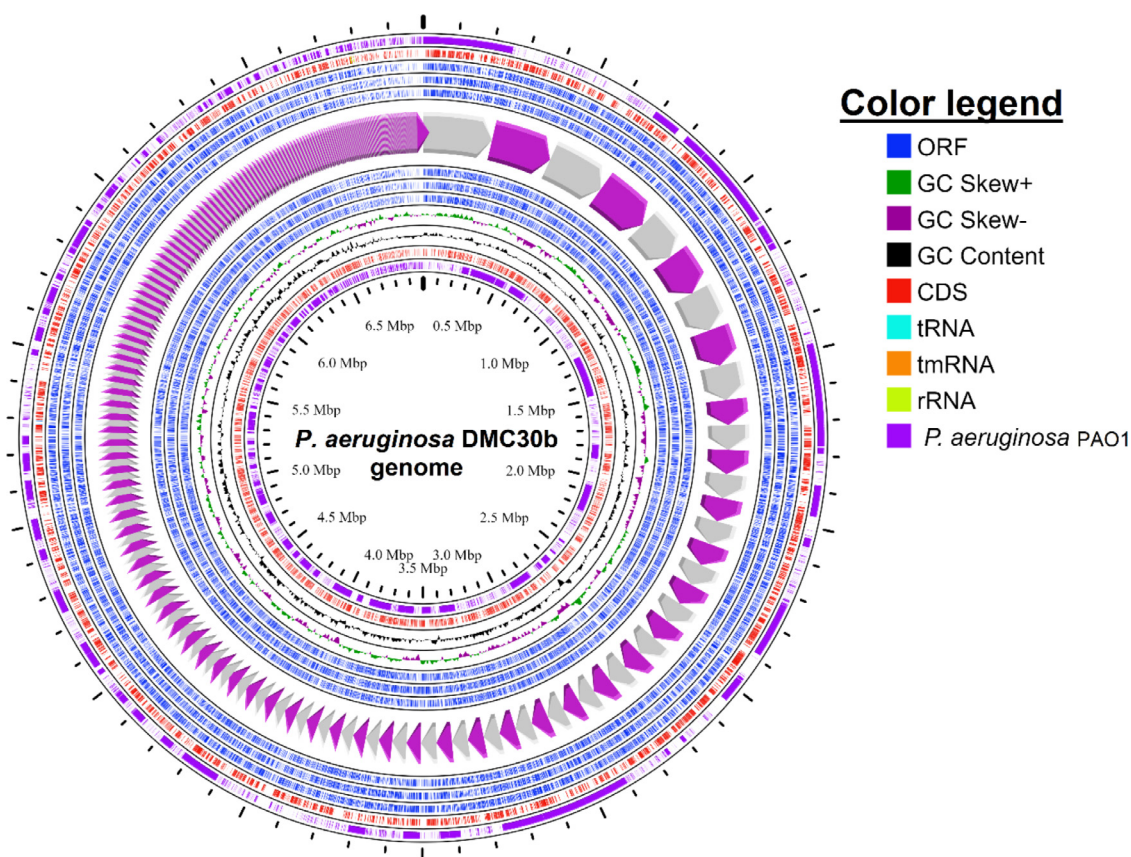


Fig. 1. Circular genome representation of *Pseudomonas aeruginosa* strain DMC30b compared with *P. aeruginosa* PAO1 (NC_002516.2). The six innermost layers in the graphic portray the genome coordinates (mega base pairs—Mbp, purple), coding sequences (CDS; red), open reading frames (ORF, blue), GC content (zigzag black line), and GC skew (green+ / deep purple – zigzag) of the *P. aeruginosa* strain DMC30b genome.

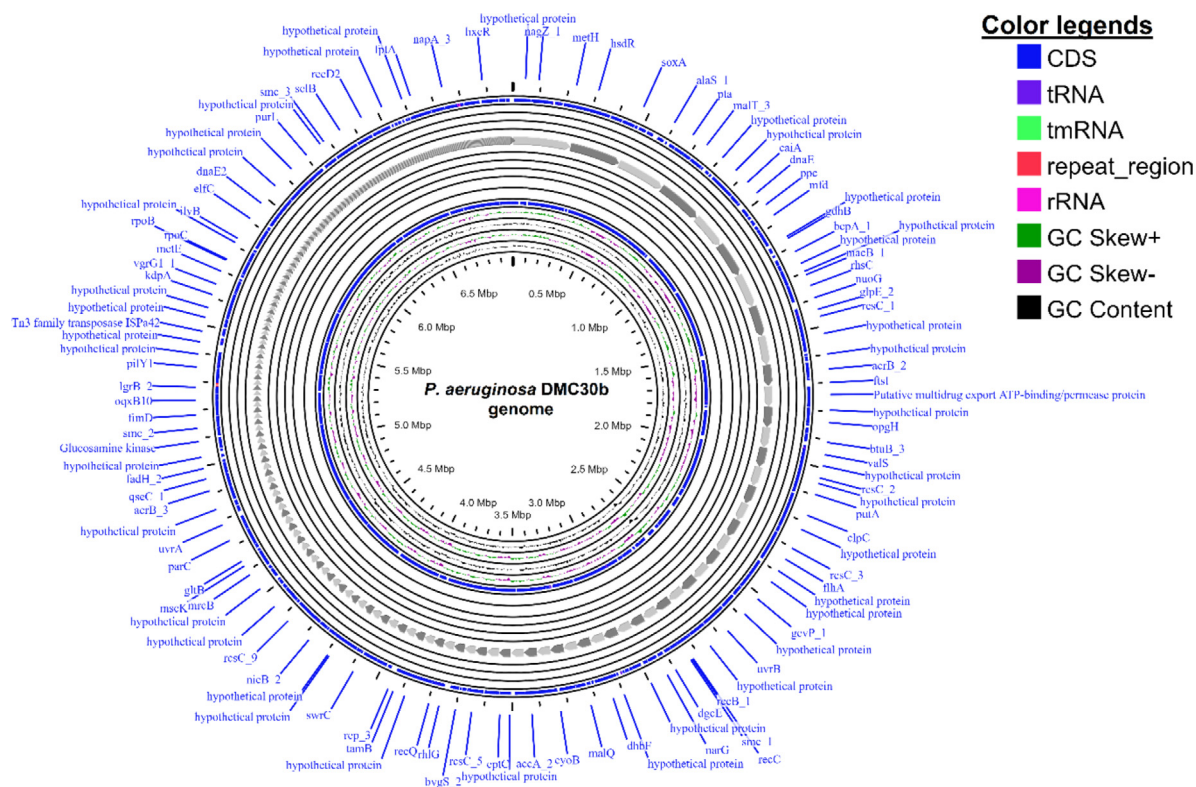


Fig. 2. Circular representation of the *Pseudomonas aeruginosa* strain DMC30b genome. The outermost ring shows the coding sequences with Prokka annotation (both strands combined) with genes encoding core and hypothetical proteins (blue).

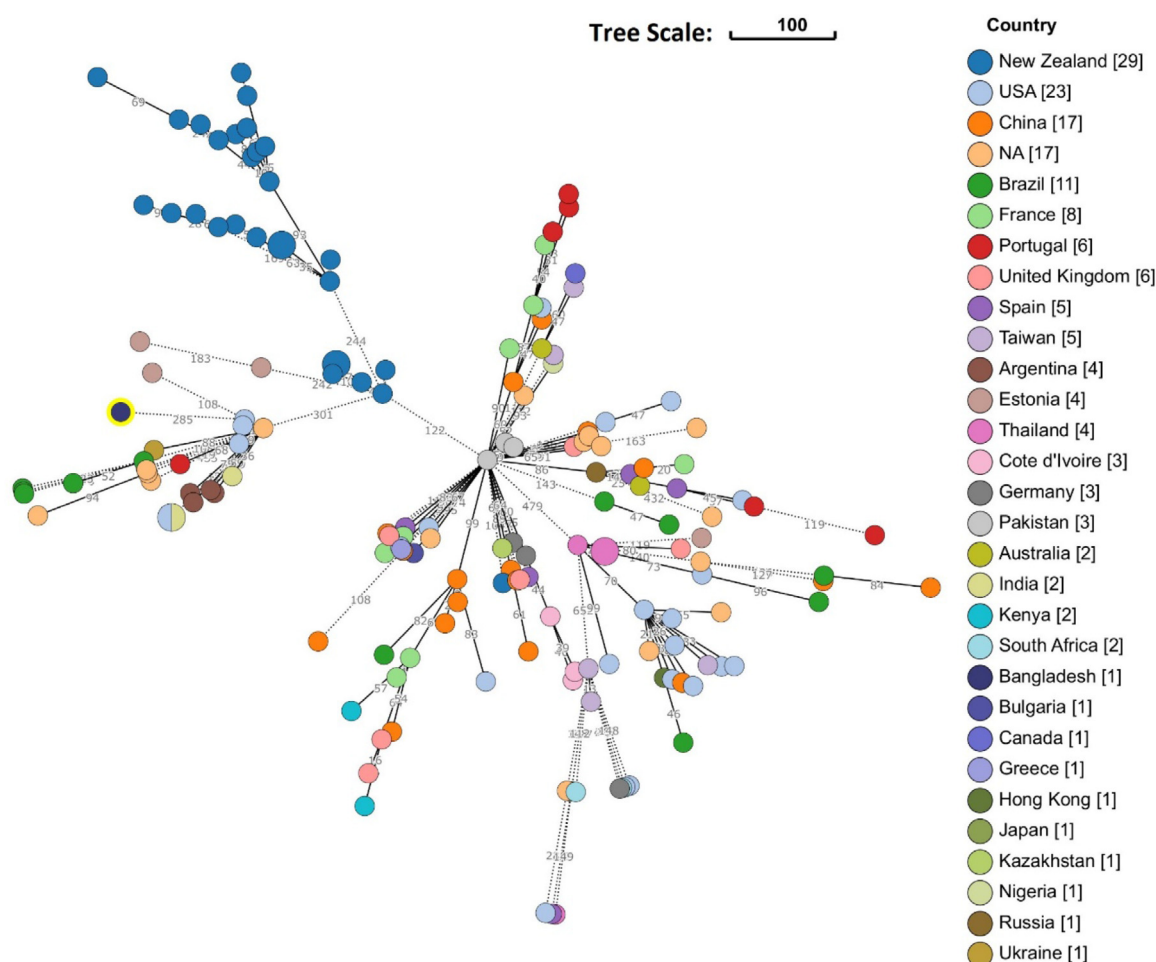


Fig. 3. Phylogenetic relationship between *Pseudomonas aeruginosa* strain DMC30b and all 166 *P. aeruginosa* strains belonging to ST244 currently available in the NCBI GenBank database by core genome multilocus sequence typing analysis. *P. aeruginosa* strain DMC30b is highlighted in yellow circle in the phylogenetic tree.

from a wide arsenal of clinical samples including enteritis, gastroenteritis (diarrheal diseases), urinary tract infections, and both community- and hospital-acquired bacteraemia in humans and animals (e.g. dogs). Horizontal gene transfer can disseminate ARG and VFG to commensal or non-pathogenic bacteria, which could lead to further infection/disease.

Two plasmid replicons were also identified in the genome of the *P. aeruginosa* DMC30b: IncP-6 (contig 20; 78 007 bp in length) (Fig. S1A) and ColRNAI (contig 154; 9359 bp in length) (Fig. S1B). Structural comparison between the IncP-6 and ColRNAI plasmids predicted in this study and similar plasmids deposited in the GenBank database showed 97.59% and 91.35% similarities, respectively. BLAST comparison showed that IncP-6 plasmid of *P. aeruginosa* DMC30b displayed 100% query coverage and 97.59% nucleotide similarity to a virulent plasmid p10265-KPC (virulence plasmid of *P. aeruginosa*) isolated from a human sample from China (accession number: KU578314). The IncP-6 plasmid is a novel carbapenem resistance plasmid in the genome of *P. aeruginosa* DMC30b, and these results corroborated previously published results for the *P. aeruginosa* strain 10265, recovered from a patient with pneumonia in a public hospital in China [33]. Moreover, IncP plasmids in ST277 isolates of *P. aeruginosa* have previously been described in China [34] and Colombia [33].

3.3. Antimicrobial resistance profile of *P. aeruginosa* strain DMC30b

P. aeruginosa DMC30b was resistant to 18 antibiotics, with imipenem and meropenem MIC₉₀s of > 512 mg/L. The *P. aeruginosa*

DMC30b was also positive for the blaOXA-48 gene. The resistomes of the *P. aeruginosa* DMC30b genome consisted of 35 genes (ARGs) conferring resistance to aminoglycosides (aadA1 and aph(3')-IIb), four classes of β -lactams (OXA-10, OXA-396, OXA-847, blaPAO, blaPME-1, TEM-116), phenicol (catB7), quinolones (crpP), fluoroquinolones (gyrA), fosfomycin (fosA), sulphonamide (sul1), tetracycline (tetD, tetG), bicyclomycin (bcr1), polymyxin (arnA), and colistin (cprR) (Fig. 4). The efflux pump conferring resistance to multiple antibiotics was found as the predominating resistance class in the genome of *P. aeruginosa* DMC30b. For instance, MexAB-OprM, MexCD-OprJ, MexEST-OprN, MexPQ-OpmE, and MexIH-OpmD are the leading efflux pump systems conferring resistance to *P. aeruginosa* DMC30b (Fig. 4).

Therefore, the findings of this study infer that the existence of resistomes (ARGs) and related resistance mechanisms in *P. aeruginosa* DMC30b, along with four classes of beta-lactam genes and efflux pump systems, may contribute to its extensive resistance to almost all antibiotics used for treatment purposes and can cause its emergence as a pandrug-resistant bacterium in Bangladesh. *P. aeruginosa* DMC30b is a particularly pernicious pathogen, as it possesses several innate defense mechanisms against antibiotics. These results are in line with our previous findings of *P. aeruginosa* DMC-27b, where the co-existence and chromosomal inheritance of all four β -lactamase classes and acquisition of multiple resistance determinants were reported [5]. This bacterium can readily acquire genetically encoded resistance determinants from other pathogens, further increasing resistance [5,6]. Clinically significant *P. aeruginosa* strains express resistance to various antimicrobial agents in-

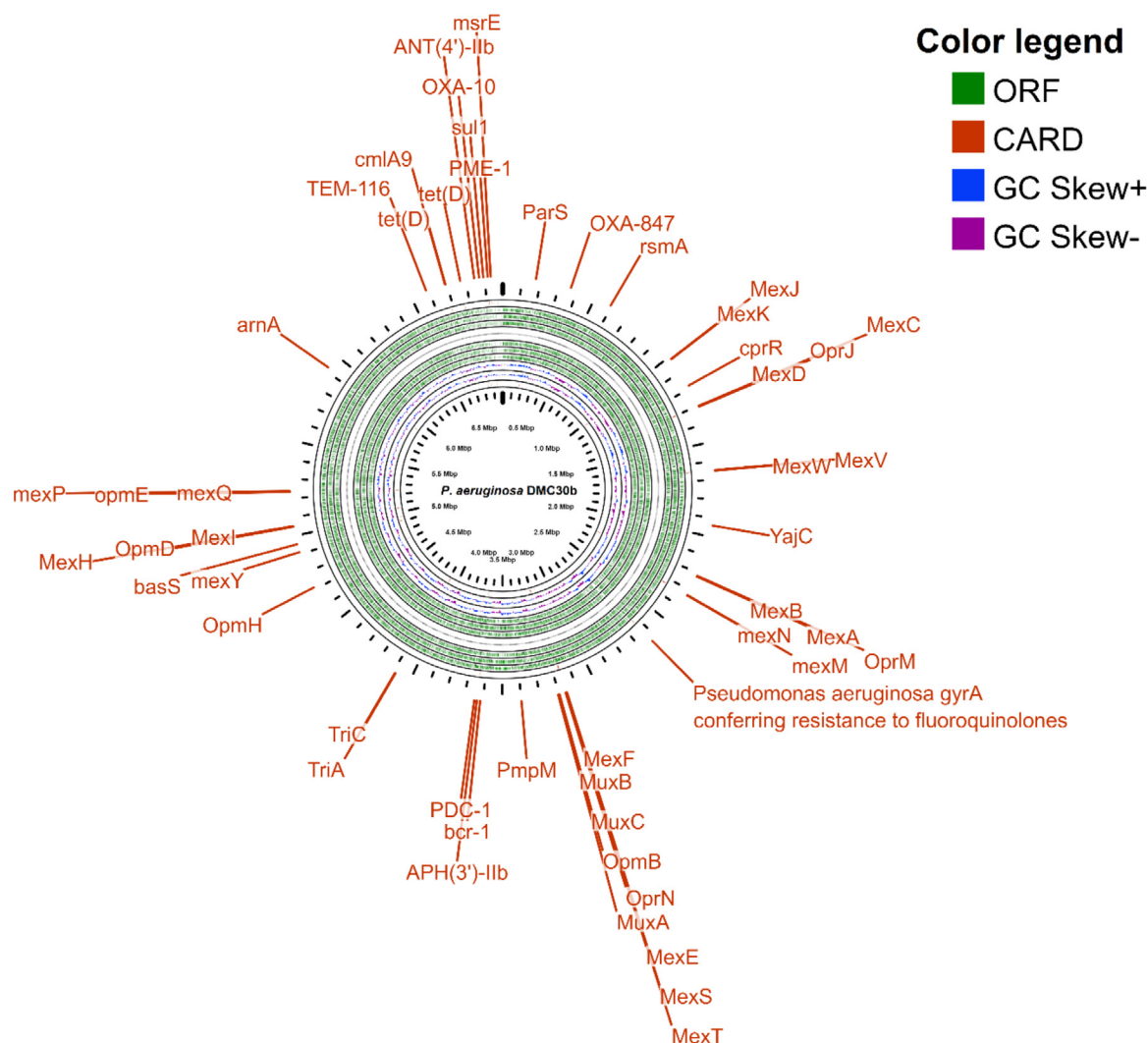


Fig. 4. Circular representation of genome *Pseudomonas aeruginosa* strain DMC30b. The outermost layer (black colour coordinates) of the circular graphic shows the antimicrobial resistances genes and related pathways (deep orange) found in DMC30b genome through Comprehensive Antibiotic Resistance Database analysis.

cluding β -lactam antibiotics through these efflux pump systems [35]. The intrinsic and acquired resistance mechanisms include tripartite efflux systems such as MexAB-OprM, MexCD-OprJ, MexEF-OprN and MexXY-OprM [35], leading to extrusion of antimicrobial agents and other xenobiotics from the cell interior. MexAB-OprM plays a role in the intrinsic resistance of *P. aeruginosa* to most β -lactams, quinolones, and many other structurally unrelated antimicrobial agents [35]. Likewise, MexEF-OprN is capable of extruding quinolones [36], and MexXY-OprM expels aminoglycosides and cephalosporin [8,35] from bacterial cell interiors. Furthermore, *P. aeruginosa* achieved high-level (MIC > 1 mg/mL) triclosan resistance either by constitutive expression of triABC, an efflux pump of the resistance nodulation cell division (RND) family, or expression of MexCD-OprJ, MexEF-OprN, and MexJK-OpmH in regulatory mutants, supporting the notion that efflux is the primary mechanism responsible for this bacterium's high intrinsic and acquired triclosan resistance [8,35]. Moreover, a series of studies reported that most of the ST244 isolates (85.3%) exhibited a MDR phenotype, i.e. being resistant to carbapenems, aminoglycosides, and fluoroquinolones [10,12].

3.4. Virulence repertoire of *P. aeruginosa* strain DMC30b

P. aeruginosa DMC30b strain harboured 214 VFGs. Among these VFGs, genes coding for phenazine biosynthesis (PhzB), flavin de-

pendent hydroxylase (PhzS), pyochelin biosynthetic protein (PchG), type VI secretion system (icmF1/tssM1, VgrG1), Fe(III)-pyochelin receptor precursor (fptA), pyoverdine biosynthesis proteins (PvdG), flagellar activities (fliA, fliL, fliO, flip, fliQ, fliR, fleN, motC, fliM etc.), twitching motility (pilU, pilG, pilH, pill, pilT and pilJ), *Pseudomonas* protein phosphatases (PppA and PppB), two-component regulatory systems (AlgB, PilR, PilS), transcriptional regulators (PchR, PtxR), and LasA protease precursors, etc were found to be enriched in the *P. aeruginosa* DMC30b genome. Pyoverdine, a siderophore produced by *P. aeruginosa*, is essential for pathogenesis in mammalian infections [37] and could represent a novel drug or vaccine target. One of the principle regulatory mechanisms for *P. aeruginosa*'s virulence is the iron-scavenging siderophore pyoverdine, as it governs in-host acquisition of iron, promotes expression of multiple virulence factors, and is directly toxic [37]. Bacterial twitching motility is a surface-associated movement commonly used by Gram-negative bacteria driven by several associated protein functions [38]. Moreover, twitching activity of *P. aeruginosa* play a role not only in motility but also in cell-cell adhesion, cell-surface adhesion and horizontal gene transfer in the pathogenesis of different diseases [39,40]. The protein phosphatases (PppA and PppB) regulate the basic cellular processes and virulence in *P. aeruginosa* [41]. *P. aeruginosa* encodes a large set of transcription factors that modulate and manage cellular metabolism to survive in variable environmental conditions, including that of the human body, and control

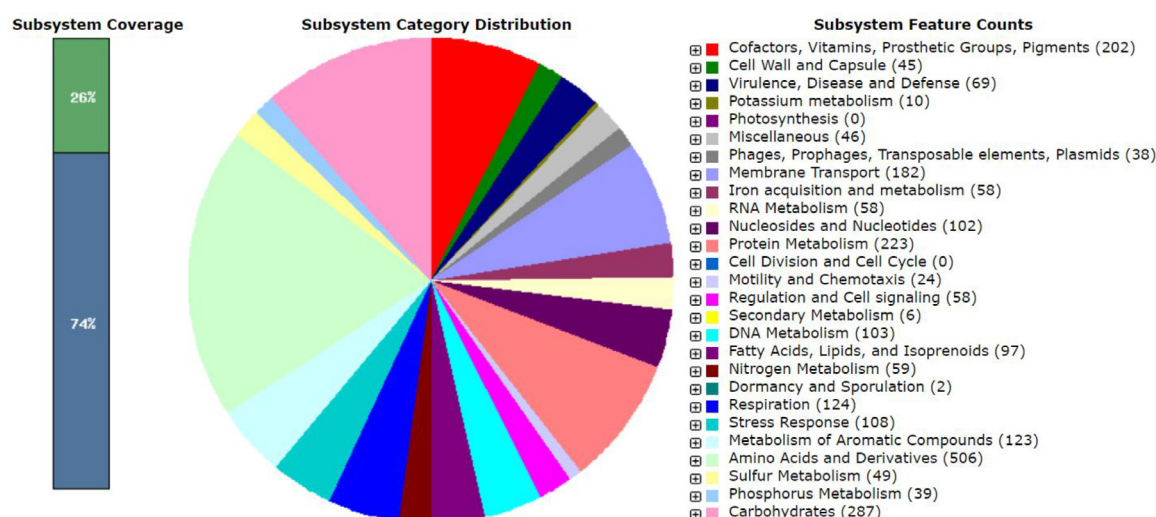


Fig. 5. An overview of the subsystem categories assigned to the metabolic genes and pathways predicted in the genome of *Pseudomonas aeruginosa* strain DMC30b. The green part in the bar-chart at the leftmost position corresponds to the percentage of proteins involved (26%) and the pie chart in the right panel demonstrates the fraction and count (parenthesis in legend) of each subsystem feature.

the expression of quorum sensing and protein secretion systems [30]. Phenazines can modify cellular redox states, regulate patterns of gene expression, enhance survival, and contribute to competitiveness, biofilm formation, and virulence in the opportunistic pathogen *P. aeruginosa* [42]. LasA is staphylolytic proteinase that is secreted by the opportunistic pathogen *P. aeruginosa*. LasA protease play a major role in the colonisation of the bacteria during bacterial keratitis by preventing other bacteria from colonisation on the cornea. For example, in the mixed infection with *S. aureus*, the enzyme eradicates the bacteria by lysis and finally eliminates the competitive bacteria for *P. aeruginosa* [10,30].

3.5. Metabolic functional potentials of *P. aeruginosa* strain DMC30b

The genome of the *P. aeruginosa* DMC30b was further searched for functional genes and pathways associated with metabolism through the RAST server. We identified a total of 340 subsystems with 26% subsystem coverage in the study genome (Fig. 5). The distribution of the genes assigned to different subsystems revealed the highest number genes encoding for amino acid metabolism (506 genes), followed by the metabolism of carbohydrates (287 genes), cofactors, vitamins, prosthetic group, and pigments (202 genes), aromatic compounds (123 genes), DNA (103 genes), and fatty acids, lipids, and isoprenoids (97 genes). We also found 182 genes related to 'membrane transport', 124 genes for 'respiration' and 108 genes for 'stress response' (Fig. 5). The *P. aeruginosa* DMC30b genome also encodes for 69 genes for virulence, disease, and defense in the subsystem analysis (Fig. 5), of which 52 (75.36%) were associated with resistance to antibiotics and toxic compounds, 14 (20.29%) for invasion and intracellular resistance, and 3 (4.34%) for antibacterial peptides.

We identified different genes and proteins from the complete genome of *P. aeruginosa* DMC30b attributable to different subsystem metabolic functions, corroborating several previous studies [10,30,37]. Our analysis provides an important expansion of metabolic functional genes and/or pathways integral to the development of infection by *P. aeruginosa*. The existence of multiple *P. aeruginosa* virulence factors and their direct metabolic regulators are linked to pathogenicity. Estimating the percentage of core metabolism genes participating in the virulence of *P. aeruginosa* DMC30b, we found that genes associated with metabolism are correlated with pathogenicity. Virulence-linked pathways in oppor-

tunistic pathogens are putative therapeutic targets that may be associated with less potential for resistance than targets in growth-essential pathways [10]. However, efficacy of virulence-linked targets may be affected by the contribution of virulence-related genes for metabolism [10]. Furthermore, given its capacity to metabolize a wide variety of substrates, it is also possible that *P. aeruginosa* DMC30b possesses greater potential for enzymatic modification, virulence, and MDR mechanisms. Therefore, the metabolic diversity, transport capabilities, and regulatory adaptability that enable *P. aeruginosa* DMC30b to thrive and compete with other microorganisms probably all contribute to its high intrinsic resistance to antibiotics. Knowledge of the complete genome sequence and encoded processes (ARGs, VFGs, and subsystems) provides a wealth of information for the discovery and exploitation of new antibiotic targets and hope for the development of more effective strategies to treat the life-threatening opportunistic infections caused by *P. aeruginosa* in humans.

4. Conclusions

This is the first WGS-based study on MDR *P. aeruginosa* DMC30b isolated from the septic wound swab of a severe burn patient in Bangladesh. In this study, we report the genomic characteristics of a MDR *P. aeruginosa* DMC30b carrying different ARGs, VFGs, and genomic functional potentials, which may help to elucidate the dissemination mechanisms of resistance and virulent genes among bacteria, animals, and humans. These data can facilitate unravelling the genomic features, antimicrobial resistance mechanisms, and epidemiological characteristics of this bacterial pathogen in the future. Further studies are required to have a wider understanding of genetic features of *P. aeruginosa* isolates circulating within communities in Bangladesh. This could ultimately bring about informed government policy and adequate community awareness toward curbing the transmission and continuous emergence of such dangerous pathogens.

5. Data availability

The WGS data of *P. aeruginosa* DMC30b are deposited at DDBJ/ENA/GenBank under accession number JAMQYG000000000 (BioSample SAMN28906490), and the assembly reports of the genome are also available at GenBank (<https://ncbi.nlm.nih.gov/>)

nucore/JAMQYG000000000). The version described in this paper is version JAMQYG000000000.1. The Ion Torrent FASTQ reads are available in the NCBI under BioProject accession number PRJNA846956.

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Ethical approval

The study was approved by the Ethical Review Committee (ERC), Faculty of Biological Sciences, University of Dhaka, Bangladesh (reference 64/Biol.Sc./2018–2019) and carried out under the direct supervision of the laboratory biosafety officer. As the samples were collected from Dhaka Medical College Hospital, we did not handle the human subjects directly.

Competing interests

None declared.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jgar.2022.08.023.

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